Cyclone Remal (2024): paper figures

This document shows the eight figures that `Remal_Paper_Figures.ipynb` produces. The notebook draws them from the outputs of the analysis notebook, so running both on the same data reproduces these images.

Preprint: Hakim, M. A. (2026). *WRF-ARW Simulation of Cyclone Remal (2024): A Comparative Assessment of FNL, GFS, CFSv2, and ERA5 Forcing Datasets.* SSRN preprint. https://doi.org/10.2139/ssrn.7609299. Captions are as in the preprint.

Figure 1

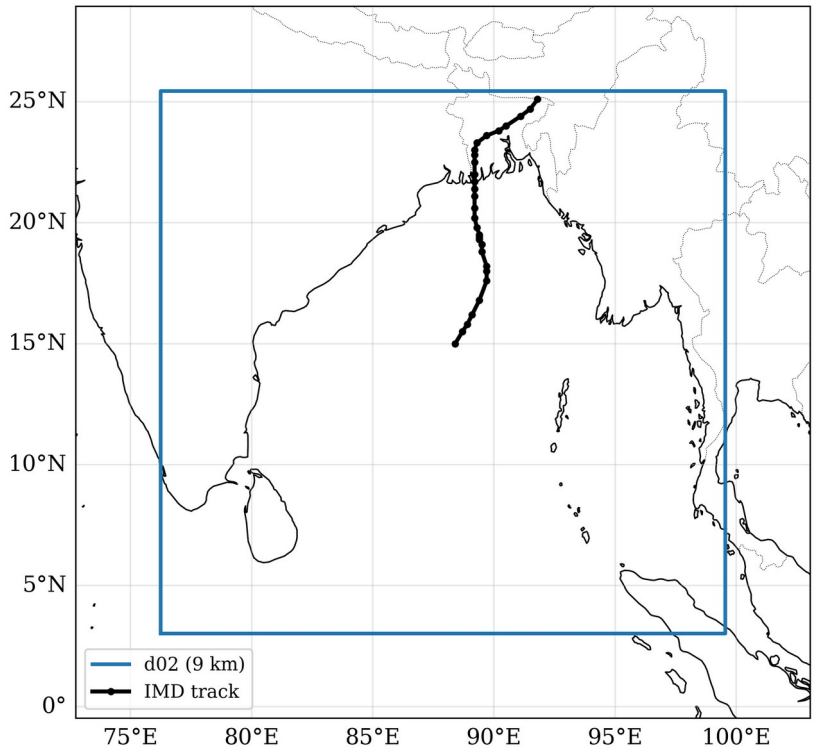


Figure 1. Inner WRF analysis domain (d02, 9 km) and the IMD best-track path of Cyclone Remal from the IMD Cyclone Warning Division report (IMD, 2024). The outer 27-km parent domain was part of the nested integration but is not shown in this diagnostic map.

Figure 2

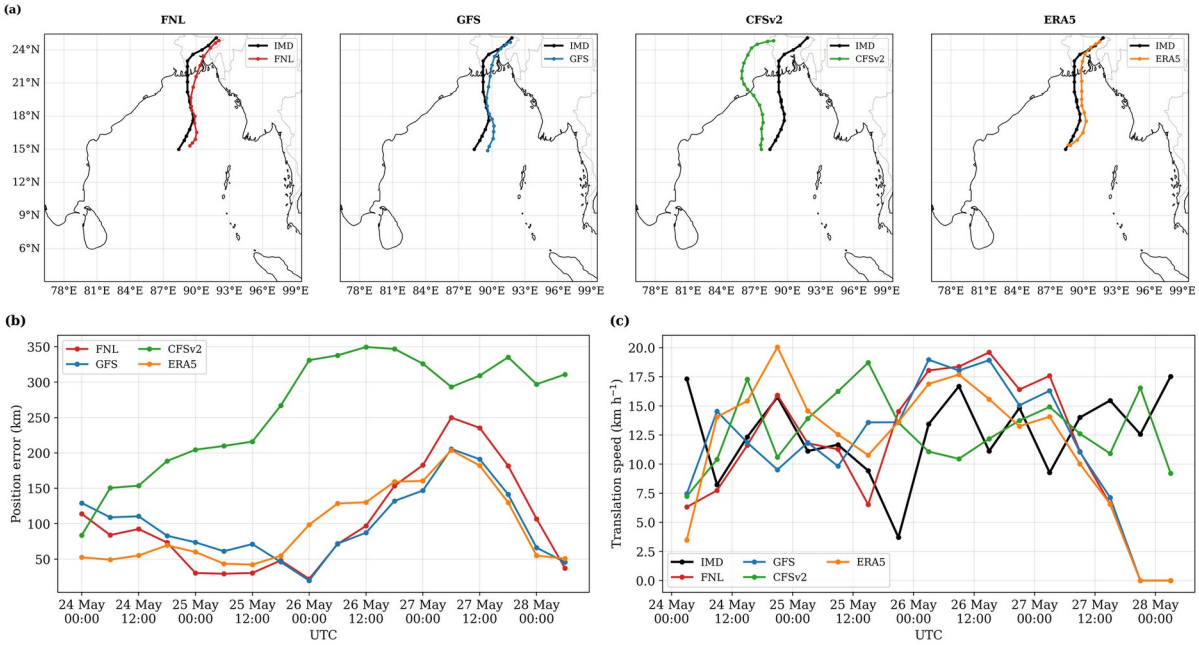


Figure 2. Track and motion validation of Cyclone Remal. (a) Simulated cyclone tracks from FNL, GFS, CFSv2, and ERA5 compared with the IMD best track; (b) 6-hourly position error relative to IMD; and (c) cyclone translation speed. Track position statistics use the frozen 6-hourly workflow; the zero translation speeds at the end of the FNL, GFS, and ERA5 series reflect the carried-forward post-landfall positions (Section 4.5).

Figure 3

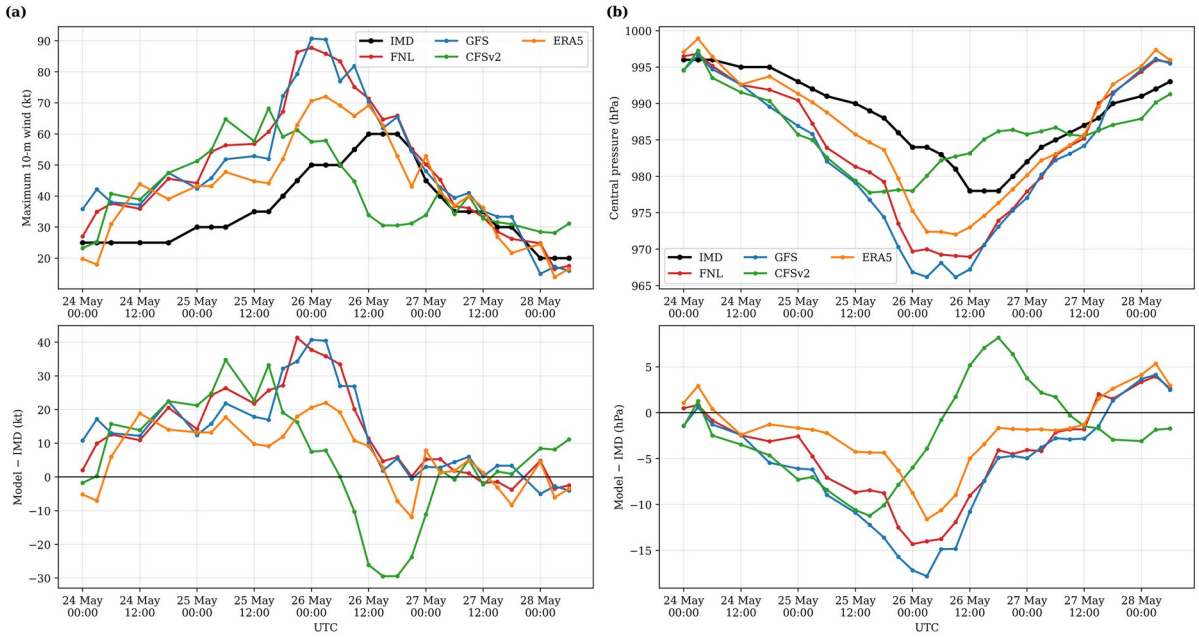


Figure 3. Intensity validation against IMD over the 30 exact common timestamps. (a) Maximum 10-m wind within 150 km of the diagnosed center and model-minus-IMD wind bias; (b) simulated central pressure and model-minus-IMD pressure bias.

Figure 4

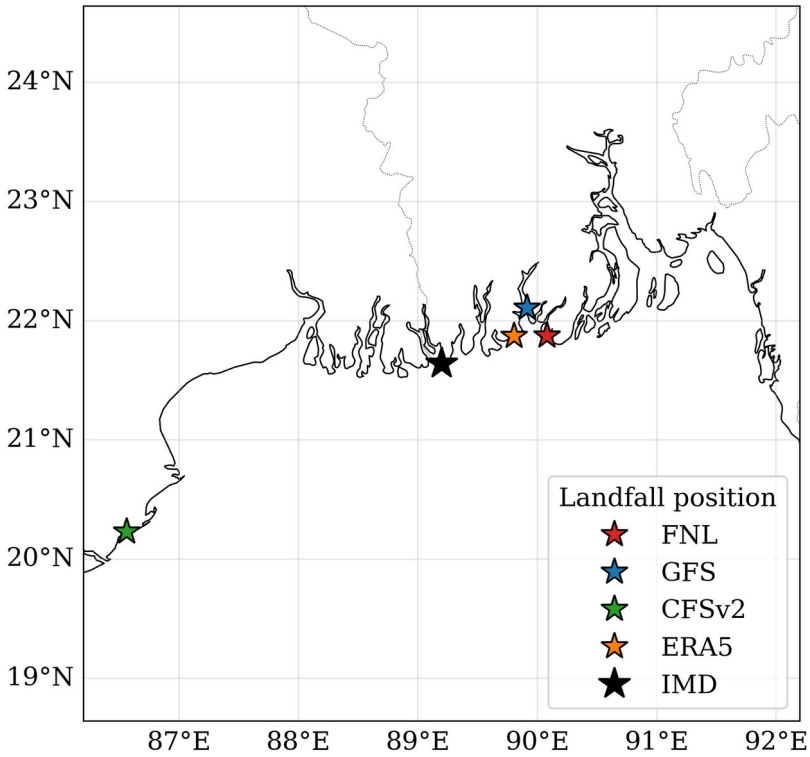


Figure 4. Observed and simulated coastal crossings. Stars mark the IMD-derived and model-derived landfall points obtained from the respective tracks and the common WRF land mask.

Figure 5

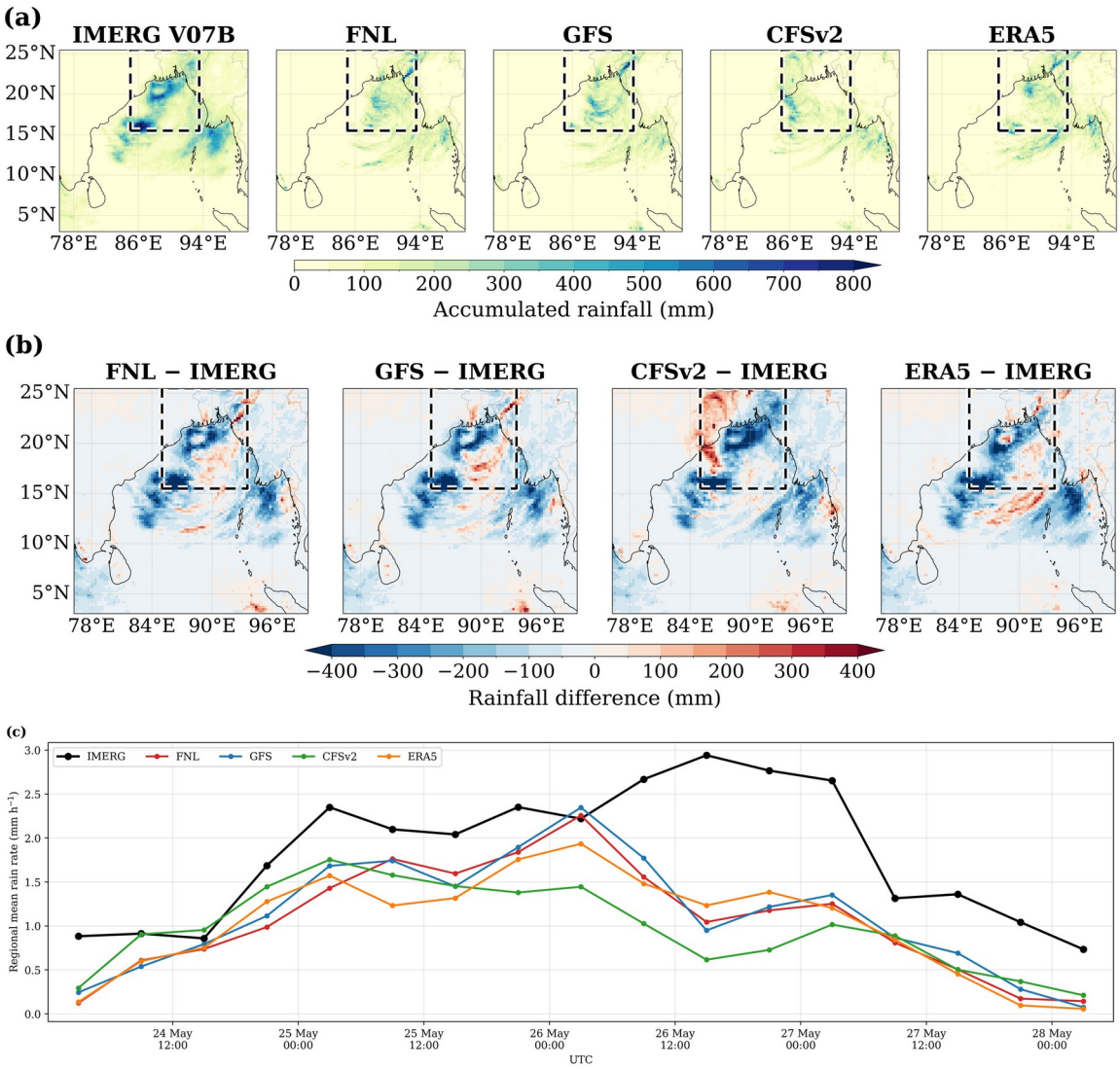


Figure 5. Rainfall validation against IMERG V07B. (a) Common-period accumulated rainfall from IMERG and the four WRF experiments; (b) WRF-minus-IMERG accumulated-rainfall differences; and (c) regional mean rainfall-rate evolution on the common 6-hourly support. The maps show the full d02 domain; the dashed box marks the 85.0–93.5°E, 15.5–25.5°N verification region in which the rainfall statistics (Table 5) and the rain rates in (c) are computed.

Figure 6

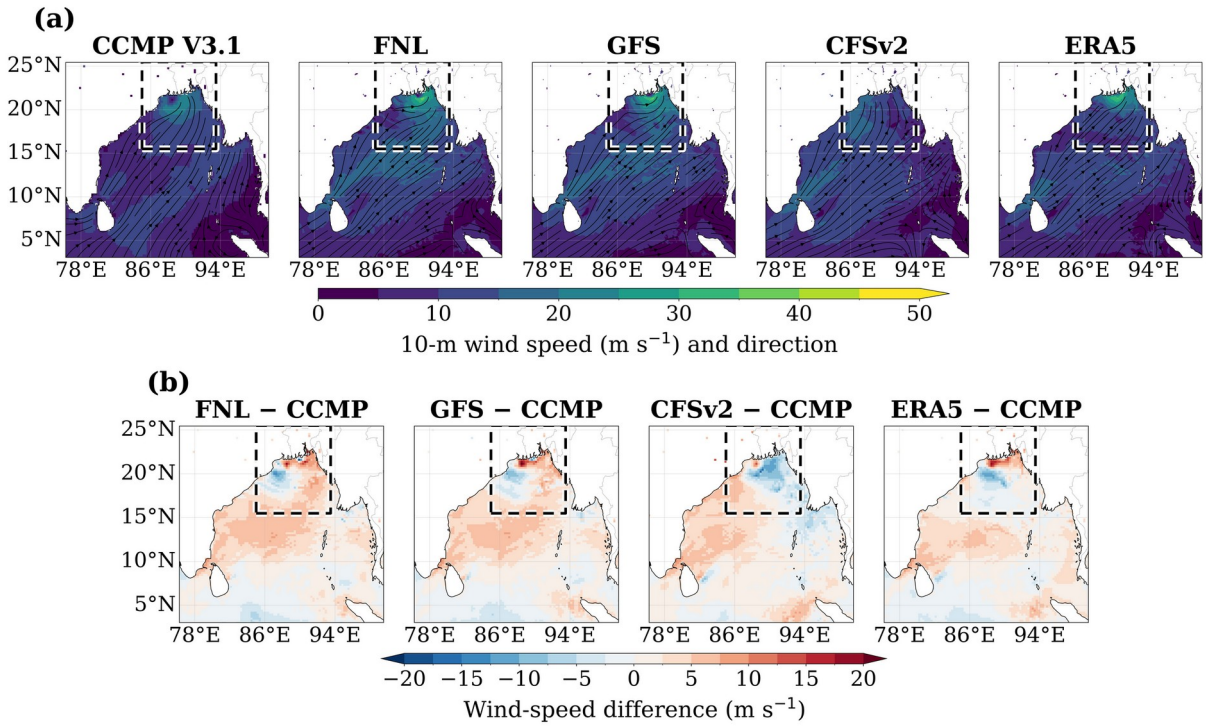


Figure 6. Ocean-surface wind validation against CCMP V3.1 at 1200 UTC 26 May 2024. (a) CCMP and WRF 10-m wind-speed fields with streamlines; (b) WRF-minus-CCMP wind-speed differences. Land points are masked because the comparison is restricted to ocean points; the dashed box marks the verification region used for the CCMP statistics in Table 5.

Figure 7

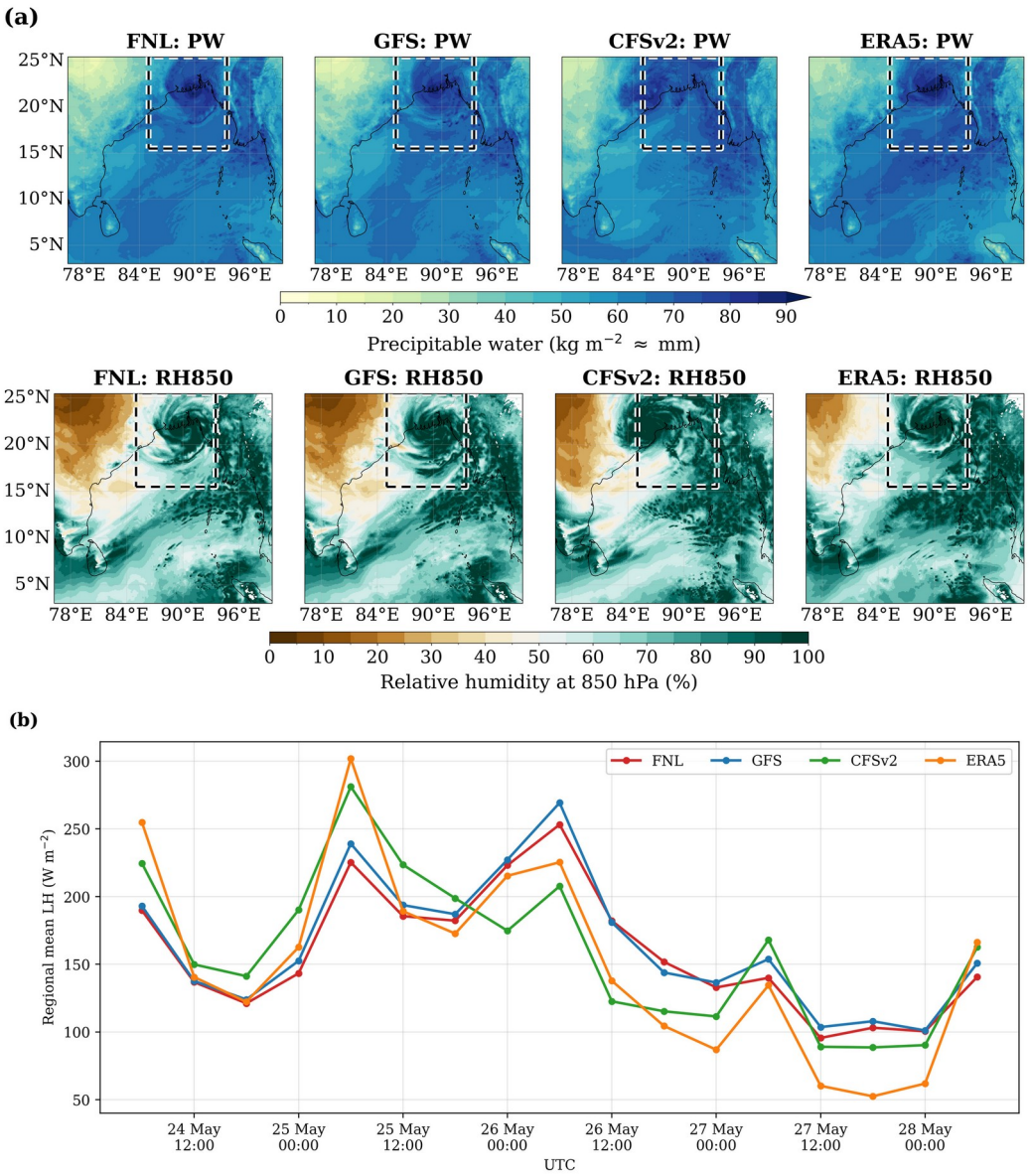


Figure 7. Thermodynamic diagnostics. (a) Precipitable water (PW) and 850-hPa relative humidity (RH_{850}) at 1200 UTC 26 May 2024; (b) WRF surface latent heat flux from the LH diagnostic (W m^{-2}), area-averaged over the dashed box in (a) (85.0–93.5°E, 15.5–25.5°N) from 0600 UTC 24 May through the analysis period. All fields are derived from the WRF experiments.

Figure 8

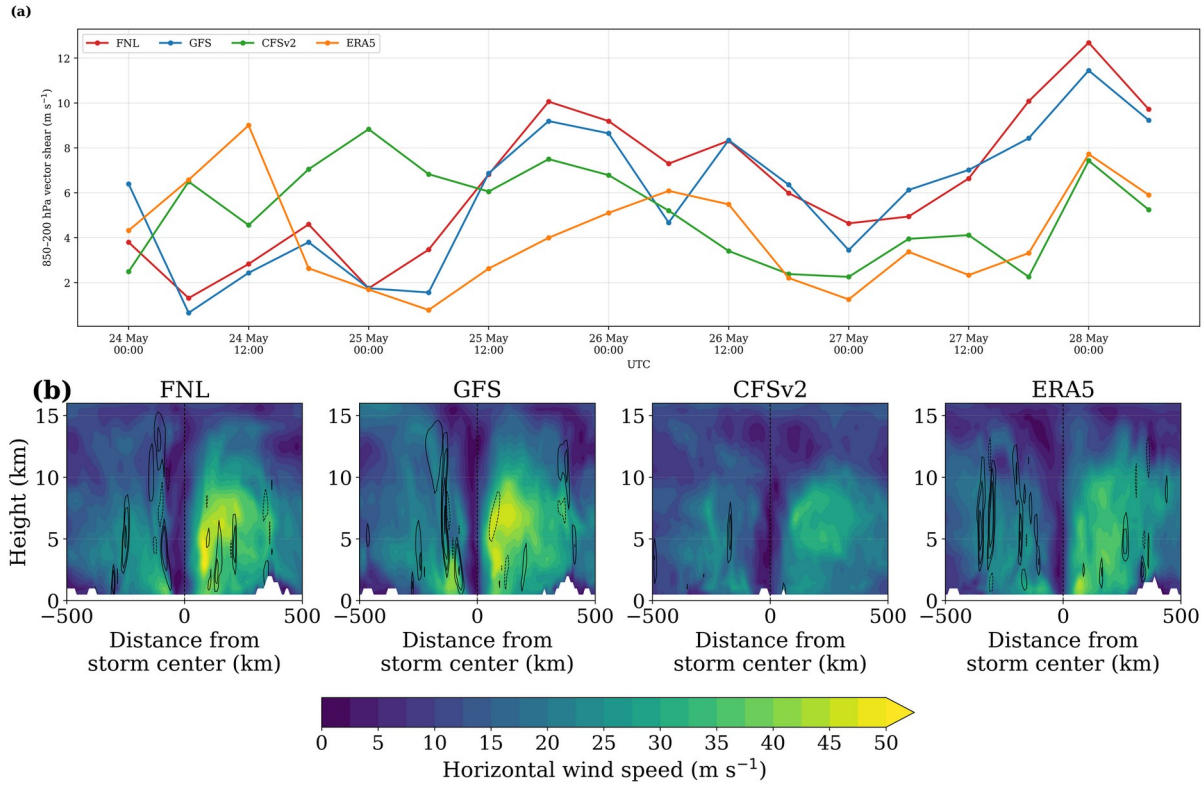


Figure 8. Environmental and storm-structure diagnostics. (a) 850–200-hPa environmental vertical wind shear averaged over the 200–500 km storm-relative annulus; (b) storm-centered east–west vertical sections at 1200 UTC 26 May 2024, with horizontal wind speed shaded and vertical velocity contoured at −1, −0.5, 0.5, 1, 2, and 4 m s^{-1} (solid, ascent; dashed, descent).